

P8131 Midterm

March 10, 2023

Name:

uni:

Please read the pledge and the instructions carefully.

Pledge: I have neither given nor received any unauthorized aid in completing this exam and I have conducted myself within the guidelines of the University Honor Code.

Signature:

1. You have 80 minutes to complete this exam.
2. This is an **open book, open note** exam.
3. There are 3 questions in the exam. Please **show all of your work and your calculations** to receive full / partial credit.
4. The exam is out of 100 points. Different questions have different weights, which are marked in front. Note that some questions might require more time than others, so make sure you manage your time accordingly.
5. **You must sign your name underneath the pledge. An exam copy without signature will not be accepted.**

Good luck!

Question 1 (30 points)

Let X be a random variable with probability density function

$$f(x, \theta) = \begin{cases} \frac{1}{\sqrt{2\pi\tau}} x^{-1} e^{-(\log x - \theta)^2 / (2\tau^2)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Here θ is the only parameter of the distribution and $\tau > 0$ is known.

1a (12 points) Show that the above distribution forms an exponential family.

1b (12 points) Determine the mean, variance, and variance function of $\log X$.

1c (6 points) What is the distribution of $\log X$? Justify your answer.

Question 2 (40 points)

Researchers would like to find out the association between prostate cancer treatment and individual-level demographic data (e.g., age, year of diagnosis, race). We have 8 age groups (i.e., 40-44, 45-49, 50-54, 55-59, 60-64, 65-69, 70-74 and 75-79, coded as a continuous variable with values 0 to 7), 3 race groups (i.e., black, white and other, where “other” is the reference group), and 13 consecutive diagnosis years (i.e., 2004 to 2016, coded as a continuous variable with values 0 to 12). A patient may have received radiation, surgery, or neither (i.e., no treatment). As a first step, we combine the “radiation” and “surgery” groups as a new treatment group and fit a logistic regression (treatment=1; no treatment=0). Each observation is corresponding to one patient. Below is a snapshot of the model fitting output from R and an estimate of the covariance matrix of the coefficients.

```
Call:
glm(formula = resp.new ~ age + year + race, family = binomial(link = "logit"))

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0500   0.5279   0.6078   0.6922   0.8726

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.23215     0.51098   4.368 1.25e-05 ***
age          -0.04994     0.05358  -0.932  0.35136
year         -0.07053     0.02306  -3.059  0.00222 **
raceBlack    -0.26679     0.43460  -0.614  0.53931
raceWhite    -0.06155     0.43352  -0.142  0.88709
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 957.77  on 978  degrees of freedom
Residual deviance: 945.52  on 974  degrees of freedom

AIC: 955.52

Number of Fisher Scoring iterations: 4

> vcov(fit)
            (Intercept)          age          year      raceBlack      raceWhite
(Intercept)  0.261098772 -1.379943e-02 -3.433387e-03 -0.179402643 -1.774723e-01
age          -0.013799430  2.870977e-03  2.598981e-05  0.001267151  5.669859e-04
year         -0.003433387  2.598981e-05  5.317796e-04 -0.000127768  9.643918e-05
raceBlack    -0.179402643  1.267151e-03 -1.277680e-04  0.188878289  1.744050e-01
raceWhite    -0.177472255  5.669859e-04  9.643918e-05  0.174405037  1.879405e-01
```

2a (5 points) Provide an interpretation of the estimated intercept of the model.

2b (5 points) How do you interpret the negative coefficient associated with variable **age**?

2c (5 points) What conclusion are you able to draw concerning the race groups?

2d (10 points) What is the probability of treatment for a *black* patient in *age group 70-74* and *diagnosis year 2011*?

2e (10 points) What is the odds ratio of treatment for a *white* patient in *age 45-49* vs a patient of neither black nor white in *age group 60-64* when both were diagnosed in the same year? What is the 95% confidence interval for this odds ratio?

- 2f (5 points) Suppose we would like to know whether the race effect on treatment differs by the diagnosis year based on deviance analysis. Describe the two models needed for this deviance analysis and specify the degrees of freedom of the chi-squared distribution used for this test.

Question 3 (30 points)

In a study of motor vehicle safety, 150 men and 150 women were interviewed to rate how important air conditioning and power steering were to them when they were buying a car.

Sex	Age	Response			Total
		Unimportant	Import	Very Import	
Women	18-23	26 (58%)	12 (27%)	7 (16%)	45
	24-40	9 (20%)	21 (47%)	15 (33%)	45
	> 40	5 (8%)	14 (23%)	41 (68%)	60
Men	18-23	40 (62%)	17 (26%)	8 (12%)	65
	24-40	17 (39%)	15 (34%)	12 (27%)	44
	> 40	8 (20%)	15 (37%)	18 (44%)	41
Total		105	94	101	300

- 3a (10 points) A proportional odds model was fit and the summary is given below.

Call:

```
polr(formula = res ~ sex + age, data = car.ord, weights = freq)
```

Coefficients:

	Value	Std. Error	t value
sexM	-0.5762	0.2262	-2.548
age24-40	1.1471	0.2776	4.132
age>40	2.2325	0.2915	7.659

Intercepts:

	Value	Std. Error	t value
unim im	0.0435	0.2323	0.1874
im veim	1.6550	0.2556	6.4744

Residual Deviance: 581.2956

According to this model, who has a higher probability to report “very important”, a 19-year-old lady or a 28-year-old gentleman? Justify your answer.

3b (10 points) According to the above model, who has a higher probability to report “important”, a 19-year-old lady or a 28-year-old gentleman? Justify your answer. Is your answer to this question the same as the one in part (a)? Why so?

3c (10 points) Ignore the ordering of the outcome, the following nominal regression model was fit and summary provided in R.

```
Call:
multinom(formula = cbind(res.unim, res.im, res.veim) ~ sex +
  age, data = car)
```

Coefficients:

	(Intercept)	sexM	age24-40	age>40
res.im	-0.5907992	-0.3881301	1.128268	1.587709
res.veim	-1.0390726	-0.8130202	1.478104	2.916757

Std. Errors:

	(Intercept)	sexM	age24-40	age>40
res.im	0.2839756	0.3005115	0.3416449	0.4028997
res.veim	0.3305014	0.3210382	0.4009256	0.4229276

Residual Deviance: 580.7022

According to this new model, who has a higher probability to report “very important”, a 19-year-old lady or a 28-year-old gentleman? And who has a higher probability to report “important”? Justify your answers.